

Assignment 1: Basic OCaml

CAS CS 320: Concepts of Programming Languages

due Thursday, September 10 by 8:00PM

You'll put your coding solutions in the file `lib/assign1.ml`. See the file `test/test_assign1.ml` for example behavior of each function. You'll submit your solutions via Gradescope, where your code will be checked against a collection of hidden tests. You must pass at least a third of the gradescope unit tests on each problem to for the assignment to be considered attempted. Your code must also follow the guidelines of the course for it to be considered attempted (i.e., it must use the course standard library, it may not use any references, arrays, mutable fields, for/while loops).

For written problems (of which there are none this week), you'll submit your solutions via Gradescope as a single PDF file. They must be hand-written and exceptionally neat, and the final answer to each problem must be clear, e.g., surrounded by a box. You must cite your sources per the instructions in the course manual.

You're required to submit solutions to every problem in this assignment to receive full credit, but we will only provide detailed feedback on **bolded** problems. Since the assignment is graded on participation, we ask that you *try at least the bolded problem with as few external aids as possible*.¹ You should think of these problems as simulated quizzes or exams. *You won't be penalized for mistakes*, so it's in your best interest to do these problems on your own; the detailed feedback we provide will generally be more useful (for you and for us) if you submit your own work.

¹You should probably do this for most problems, but it is up to you how you want to approach the unbolded exercises.

OCaml Exercises

These problems are taken from the OCaml Exercises linked on the course web-page.

(1.3) Number of Digits

(1.10) Perfect Powers (*Optional*)

(1.13) Number of Factors

(1.16) Hypotenuse

(2.3) Drop Leading

Written Problems

There are no written problems this week.